

Ex-Ante ERW Model — Living Documentation

Geospatial profitability model for Enhanced Rock Weathering with basalt in Sub-Saharan Africa

2026-05-15

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Ex-Ante ERW Model — Living Documentation

Geospatial profitability model for Enhanced Rock Weathering with basalt in Sub-Saharan Africa. This document is the single source of truth for the *current* state of the model. Update it whenever anything material changes — the **Status at a glance** section, the **Changelog**, the affected block under §6, and the affected output under §7.

Companion documents that this file points to (do not duplicate):

File	Purpose
docs/erw-review.md	Evidence review (literature on ERW chemistry, feedstocks, trials, agronomy, risks).
docs/erw-parameters.md	Catalogue of every constant and assumption with citations and units.
docs/erw-README.md	Runbook: file inventory, run order, output schema, known portability issues.
docs/erw-conceptual-framework.{svg,pdf,tex}	Scientific framework figure — three streams converging on the gross-margin equation.
docs/erw-pipeline-flow.{svg,pdf,tex}	Pipeline architecture figure — how the code implements the framework.
docs/maps/*.png	Rendered indicator and outcome maps (12 currently).
flows.html	Browser-rendered view of this whole thing.

0. Status at a glance

Last updated: 2026-05-15 **Geographic scope:** Sub-Saharan Africa, 47 countries, GADM v4 boundaries **Working grid:** SoilGrids-Africa 30 arc-sec aggregated 10× ~0.083° (≈9 km) — 746 × 828 cells **Currency:** USD; **CO₂ in tonnes; year-1 / NPV @ 10% / equilibrium regimes**

Pipeline-block status

Block	Stage(s)	Status	Notes
B1. Working grid	erw-1, erw-2	REAL	SoilGrids properties + SPAM v2 (23 crops) co-registered on the 10× grid

Block	Stage(s)	Status	Notes
B2. Basalt rate	erw-3	REAL	LiTAS / Kamprath / Cochrane / maintenance + uniform 10/20/50 t·ha ⁻¹
B3. CDR surface	erw-9	REAL	Climate × pH × pedogenic-C deduction; CDR-per-tonne + CDR-per-ha rasters
B4. Logistics	erw-basalt-access	REAL (since 2026-05-15)	Friction surface fetched, mask expanded to vb+pb, end-to-end delivered price computed
B5. Energy	erw-energy-country	REAL	Country tariff + grid CI rasters
B6. Crop response — EcoCrop	erw-4, erw-5	REAL	Per-crop pH and acidity-saturation suitability rasters for 23 crops
B6. Crop response — Bayesian	erw-yield-fit, erw-yield-predict	NOT RUN	Trial-anchored brms model not fit; erw-7 falls back to EcoCrop
B7. Profitability	erw-7	NOT RUN	All inputs are ready; gross-margin rasters not yet generated
B8. Supply ranking	erw-supply	NOT RUN	Depends on B7
B9. Sensitivity	erw-8	NOT RUN	Depends on B7

Headline numbers (current run)

Metric	Value	Source raster
Mafic source pixels (vb+pb in SSA)	1,399,005 (1.9% of SSA cropland-context cells)	basalt_source_mask.tif
Mean delivered basalt price	\$40.8 / t (range \$10 – \$50, cap binds for ~half of pixels)	basalt_delivered_price_usd_t.tif
Mean travel time to source	815 min (capped at 1000 min ≈ 1000 km @ 60 km/h)	basalt_travelttime_min.tif
CDR per tonne basalt	computed pixel-wise	cdr_per_t_basalt.tif
CDR yield (LiTAS year-1)	computed pixel-wise; strongest in humid tropics	cdr_yield_merlos.tif

1. Changelog

Reverse chronological. One bullet per material change. Reference the block(s) affected so it is obvious which section under §6 also moved.

2026-05-15 — B4 logistics first end-to-end run; feedstock filter extended

- **B4** Fetched MAP motorised friction surface (2019), 72.5 MB SSA-clipped, via malariaAtlas::getRaster(dataset_id = "Explorer__2020_motorized_friction_surface"). Wrote data/access/friction_surface_2019.tif.
- **B4** Ran erw/erw-basalt-access.R end-to-end. Mean delivered price \$40.8 / t, mean travel time 815 min. Outputs: basalt_travelttime_min.tif, basalt_transport_km.tif, basalt_transport_cost_usd_t.tif, basalt_delivered_price_usd_t.tif.

- **Fix** `erw/erw-basalt-access.R:127`: original call passed the basalt mask as `target=` to `terra::costDist`; target must be the *value* in `x` that marks sources, not a separate raster. Now writes a sentinel `-1` into a copy of friction at source cells and calls `costDist(friction_src, target = -1)`.
- **Fix** `malariaAtlas 1.6.0` deprecated the `surface=` argument of `getRaster()`; new helper `erw/erw-fetch-friction-surface.R` resolves the `dataset_id` from `listRaster()`, forces EPSG:4326, and converts SSA to `sf` (the package's internal `vect()` dispatch does not accept `SpatVector`).
- **B2 feedstock** Extended GLiM filter from `c('vb')` to `c('vb', 'py', 'pb')` in both `erw/erw-build-basalt-mask.R` and `erw/erw-basalt-access.R`. SSA polygons 5,266 $\square$ 6,455; source-mask cells 1,321,857 $\square$ 1,399,005. New categorical raster `data/basalt_source_class.tif` (1 = basalt s.s., 3 = gabbro; pyroclastics absent in SSA at GLiM v1 resolution).
- **Maps** Added `docs/maps/10-basalt-traveltime.png`, `11-basalt-delivered-price.png`, `12-feedstock-class.png`. Refreshed `09-basalt-sources.png`. Gallery now 12 maps.

2026-05-14 — Conceptual framework + pipeline figures

- **Docs** TikZ-rendered figures: `docs/erw-pipeline-flow.{tex,pdf,svg,png}` (code architecture) and `docs/erw-conceptual-framework.{tex,pdf,svg,png}` (scientific framework).
- **Docs** `flows.html` now embeds both figures + 9-map gallery (later expanded to 12).
- **Pipeline figure** Rewrote arrows with full orthogonal routing — one horizontal track per arrow, no co-linear overlap, all corners 90°. Widened `erw-7` box so all six surfaces drop straight down; FAOSTAT wraps around the right edge.
- **B4 sub-mask** Built `data/basalt_source_mask.tif` from `data/LiMW_GIS 2015.gdb` via `erw/erw-build-basalt-mask.R` (the script had been writing an empty stub). Reprojection Eckert IV $\square$ WGS84 lat/lon was the unblocker.

2. Scope and units

- **Geographic extent**: 47 Sub-Saharan African countries per GADM v4, clipped to the union of GADM country polygons. Northern boundary at the southern edge of the Sahara (excludes Morocco, Algeria, Tunisia, Libya, Egypt). Madagascar included.
 - **Working resolution**: $\sim 0.083^\circ \approx 9$ km at the equator. This is the SoilGrids-Africa 30 arc-sec grid aggregated 10 $\times$ by mean. All rasters downstream of `erw-1` are co-registered on this grid.
 - **Crop coverage**: 23 SPAM v2 crops — MAIZ SORG BEAN CHIC LENT GROU SOYB COWP RICE WHEA CASS POTA SWPO YAMS BARL MILL VEGE TROF TEMF OPUL OOIL OFIB OCER (the SPAM v2 codes).
 - **Temporal regimes**: year-1, NPV at 10% over 10 years, equilibrium (saturating CDR). CDR phasing 30 / 25 / 20 / 15 / 10 % over five years (Lewis 2021).
 - **Currency**: USD throughout. Prices from FAOSTAT 2016–2020 producer-price averages, industrial-electricity tariffs from IEA / Ember / GET.invest (2022–2024).
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3. Scientific framework

See docs/erw-conceptual-framework.svg for the figure. In words:

Three causal streams converge on a per-pixel gross margin

$$\pi(x, y \mid \text{regime, allocation}) = R_{\text{CDR}}(x, y) + R_{\text{agro}}(x, y) - C(x, y).$$

3.1 Biogeochemical stream $\square R_{\text{CDR}}$

feedstock $\times$ grain size $\rightarrow$ reactive surface $\rightarrow$ climate $\times$ pH dissolution kinetics
 $\rightarrow$ net alkalinity export (minus pedogenic-C precipitation) $\rightarrow$ carbon revenue

- Feedstock chemistry sets CaO + MgO mass fractions (default 10% + 7% by mass).
- Grain size sets specific surface area; the SSA exponent and Streffer-2018 grinding-energy exponent are the two grain-size dials.
- Dissolution rate is the temperature $\times$ moisture climate factor (Arrhenius anchored to $T_{\text{ref}} = 11^\circ\text{C}$, $P_{\text{ref}} = 1000\text{ mm}$) times a triangular pH-acid-catalysis term peaked near pH 5.
- The pedogenic-C deduction (UNEP 1992 aridity bands keyed on MAP) removes alkalinity that precipitates locally instead of exporting as ocean HCO_3^- .
- Carbon revenue: $R_{\text{CDR}} = \text{CDR}_t \times (p_{\text{CO}_2} - \text{MRV})$.

3.2 Agronomic stream $\square R_{\text{agro}}$

soil acidity constraint $\rightarrow$ lime-equivalent demand $\rightarrow$ basalt application rate $\rightarrow$ crop response $\rightarrow$ yield revenue

- Soil acidity (pH, exchangeable acidity, Al saturation, ECEC) drives the lime-equivalent demand via `limer::limeRate` (Kamprath / Cochrane / Aramburu-Merlos LiTAS / maintenance).
- The lime demand is converted to t basalt ha^{-1} using the feedstock chemistry, a Lewis-2021-calibrated reactive fraction, and the grain-size lever — **this rate is the cross-link feeding the biogeochemical stream** (see the dashed arrow in the framework figure).
- Crop response per pixel is the maximum of two surfaces:
 - **EcoCrop** Recocrop suitability clamped to [0.2, 1] for pH and acidity-saturation.
 - **Bayesian uplift** posterior medians from a brms model on the trial dataset, with covariates (pH, MAT, MAP, grain, crop type). When the Bayesian surface is present erw-7 prefers it; otherwise falls back to EcoCrop.
- Yield revenue: $R_{\text{agro}} = \Delta y(x) \times p_{\text{crop}}$.

3.3 Economic-geography stream $\square C$

feedstock supply $\rightarrow$ logistics $\rightarrow$ grinding (cost + LCA) $\rightarrow$ operations $\rightarrow$ total cost

- Feedstock supply: quarry-gate fines at \$10 / t, anchored on GLiM v1 basic-volcanic, pyroclastic, and basic-plutonic source rocks.
- Logistics: cost-distance over the MAP 2019 motorised-friction surface, monetised at \$0.04 / min / t (30-t truck @ \$80 / h). Haul capped at 1000 km.
- Grinding: kWh / t $\times$ country industrial-electricity tariff $\times$ grid carbon intensity (the CI term is the LCA deduction from gross CDR, not a cost).
- Operations: \$8 / t spreading + \$30 / tCO₂ MRV (sampling + lab + verification + registry).

3.4 Decision variables

Six dials are exposed at the framework boundary and propagate inwards:

1. Feedstock chemistry — CaO + MgO mass fractions.
 2. Grain size — 10 – 500 μm .
 3. Allocation rule — LiTAS-targeted vs uniform 10 / 20 / 50 t ha⁻¹.
 4. Carbon price — default \$150 / tCO₂.
 5. MRV cost — default \$30 / tCO₂.
 6. Discount rate — default 10 % (NPV regime only).
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4. Pipeline architecture

See docs/erw-pipeline-flow.svg for the figure. In words:

- **Data layer** SoilGrids · SPAM v2 · WorldClim · GLiM + MAP friction · EcoCrop + ERW trials · FAOSTAT + IEA/Ember.
 - **Working-grid layer** erw-1, erw-2 produce soilgrids_properties_cropland.tif and spam*_processed.tif; erw-4 builds the per-crop parameter tables, erw-5 the EcoCrop response surface.
 - **Surface layer** Six geospatial surfaces:
 - erw-3 □ basalt rate
 - erw-9 □ CDR yield (per-t and per-ha)
 - erw-basalt-access □ delivered basalt price
 - erw-energy-country □ tariff + CI
 - erw-yield-fit □ posterior fit (Bayesian)
 - erw-yield-predict □ posterior surfaces (Bayesian)
 - **Convergence** erw-7 decomposed-cost profitability function; six straight drops from the surface layer plus FAOSTAT crop prices wrapping in from the right.
 - **Output layer** erw-8 sensitivity sweeps + erw-supply deployment frontier (LP-dual ranking on \$ per tonne basalt).
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5. Data sources

5.1 On disk

Located under data/. File sizes reflect SSA-clipped rasters at the working resolution.

File	Size	Source & notes
gadm_ssa.gpkg	~30 MB	GADM v4 — 47 SSA countries
soilgrids_properties_all.tif	~600 MB	SoilGrids-Africa (geodata::soil_af); 10 bands: SBD, ph, hp, k, ca, mg, na, bases, ecec, hp_sat
soilgrids_properties_cropland.tif	190 MB	as above, masked to QED cropland; cropland-only working stack
spam*_processed.tif (harv_area, prod, yield)	~70 MB each	SPAM v2 via geodata::crop_spam; 23 bands, one per crop
basalt_*.tif (kamprath, cochrane, merlos, merlos_maintenance)	up to ~270 MB	computed by erw-3; t ha ⁻¹ at the four LiTAS regimes

File	Size	Source & notes
basalt_uniform_{10,20,50}.tif	~7 MB each	computed by erw-3; uniform-rate counterfactuals
cdr_{climate_factor,pedogenic_fractallion,per_t_basalt}.tif	small	computed by erw-9; sub-components of CDR
cdr_yield_*.tif	small	computed by erw-9; per-ha cumulative CDR
electricity_usd_kWh.tif, grid_CI_kg_per_kWh.tif	small	computed by erw-energy-country; tariff + grid CI
basalt_source_mask.tif	~115 kB	binary, vb + pb; built by erw-build-basalt-mask then erw-basalt-access
basalt_source_class.tif	~110 kB	categorical (1 = basalt, 3 = gabbro); built by erw-build-basalt-mask
basalt_{traveltime_min, transport_km, transport_cost_usd_t, delivered_price_usd_t}.tif	~400 kB each	B4 cost surfaces; computed by erw-basalt-access
data/access/friction_surface_20192.5.tif	~2.5 MB	MAP motorised friction surface (2019), via malariaAtlas, ~30 arc-sec
data/LiMW_GIS_2015.gdb	~50 MB	PANGAEA doi:10.1594/PANGAEA.788537 — GLiM v1 global lithology
data/ecocrop_f/{hp,ph}_crop_suitability_*.tif	small	computed by erw-5; 23 crops × 2 responses

5.2 External datasets (provenance)

Dataset	Version	Access	Used by
GADM	v4.1	geodata::world()	erw-1
QED cropland	latest	geodata::cropland(source='QED')	erw-1
SoilGrids-Africa	iSDA 30 arc-sec	geodata::soil_af()	erw-1
SPAM	v2 (Africa)	geodata::crop_spam()	erw-2
EcoCrop parameter set	curated Re-cocrop::ecocropPars()	local CSV	erw-4
WorldClim	v2.1 (MAT, MAP)	geodata::worldclim_global	erw-9, erw-yield-predict
GLiM lithology	v1.0 / 1.1 (GDB)	PANGAEA download	erw-basalt-access
MAP motorised friction surface	nominal 2019	malariaAtlas (dataset_id Ex-plorer_2020_motorized_friction_surface)	erw-basalt-access
FAOSTAT producer prices	2016 – 2020 average	manual CSV download	erw-7, erw-8
Country energy (industrial tariff + grid CI)	2022 – 2024	hard-coded CSV in erw-energy-country.R	erw-energy-country
ERW trial dataset	curated by Bisrat	erw/erw-trial-data.csv (~14 rows)	erw-yield-fit

5.3 Not represented (gaps)

- **Industrial alkaline by-products:** steel-slag stockpiles (Saldanha and Vanderbijlpark in South Africa, Ajaokuta in Nigeria, Helwan in Egypt — outside SSA but referenced), cement-kiln dust at clinker plants, mine tailings at copper/nickel operations. These would add 10^5 – 10^6 tonnes / yr of point-source feedstock supply in countries with no basalt outcrops (notably much of West Africa).

- **Ultramafic outcrops:** olivine, dunite, peridotite, serpentinite. Intentionally excluded for crop-land deployment because of Ni / Cr leaching and asbestos risk (see review §3).
- **National geological maps:** some SSA countries (Tanzania, South Africa) have higher- resolution geological mapping than GLiM v1. Not currently integrated.

6. Methodological blocks

Each block is one functional unit of the conceptual framework. The script(s) that implement it are listed; the **Status** mirrors §0; the **Where to look** column points to the file the reader edits when the block changes.

B1. Working grid — erw-1, erw-2 — REAL

GADM v4 SSA mask; depth-weighted SoilGrids properties (pH, BD, exch. acidity, K, Ca, Mg, Na bases, ECEC, acidity-saturation); SPAM v2 harvested-area / production / yield for 23 crops resampled to the 10× working grid. ECEC and acidity-saturation surfaces derived inline.

Where to look: erw/erw-1-layers-soilgrids.R, erw/erw-2-layers-spam.R, docs/erw-parameters.md §1 (depth-weighting recipe).

B2. Basalt rate — erw-3 — REAL

Four lime-equivalent CaCO_3 rates (Kamprath, Cochrane, Aramburu-Merlos LiTAS year-1, LiTAS maintenance) via `limer::limeRate`. Converted to t basalt ha^{-1} using user-set CaO/MgO mass fractions (default 10% / 7%), the Lewis-2021-calibrated reactive fraction, and the grain-size lever. Uniform 10 / 20 / 50 t ha^{-1} counterfactuals computed in parallel.

Key constants

Name	Default	Notes
CaO_frac	0.10	Mass fraction of CaO in feedstock.
MgO_frac	0.07	Mass fraction of MgO in feedstock.
grain_size_um	100 μm	Sweeps 10 – 500 μm in erw-8.
reactive_fraction_ref	0.20	Lewis 2021 calibration at reference grain × climate.
GRAIN_BETA	0.5	Surface-area exponent for the grain-size factor.
GRIND_ALPHA	1.2	Strefler 2018 grinding-energy exponent.

Where to look: erw/erw-3-basalt-requirements.R, docs/erw-parameters.md §2.

B3. CDR surface — erw-9 — REAL

Per-tonne CDR potential from CaO/MgO stoichiometry, scaled by a temperature × moisture climate factor (anchored to $T_{re}f = 11\text{ }^\circ\text{C}$, $P_{re}f = 1000\text{ mm}$) and a triangular pH term peaked near pH 5. An aridity-binned pedogenic-carbonate fraction (UNEP 1992 bands on MAP) removes alkalinity precipitating locally.

Key constants

Name	Default	Notes
T_ref	11 °C	Reference MAT anchor (US Corn Belt).
P_ref	1000 mm	Reference MAP anchor.
f_pH	triangular peak at pH $\approx$ 5.0	Acid-catalysed dissolution scaling.
pedogenic_frac	MAP-binned 0 – 90 %	UNEP 1992 aridity bands using MAP as v1 proxy for AI = P / PET.

Where to look: `erw/erw-9-cdr-yield.R`, `docs/erw-parameters.md` §3.

B4. Logistics — erw-basalt-access — REAL (since 2026-05-15)

Basalt sources from GLiM v1 (vb basic volcanic, py pyroclastic, pb basic plutonic); cost-distance from each cropland pixel to nearest source over the MAP 2019 motorised friction surface; monetised at \$0.04 / min / t (30-t truck @ \$80 / h); \$10 / t quarry-gate.

Current outputs

- Source-mask cells in SSA: **1,399,005** (vb 1,320,820 + pb 78,185; py absent in SSA)
- Mean delivered price: **\$40.8 / t**, range \$10 – \$50 / t (cap binds for ~half of pixels)
- Mean travel time: **815 min**, capped at 1000 min ($\approx$ 1000 km @ 60 km/h free-flow proxy)

Key constants

Name	Default	Notes
QUARRY_GATE_PRICE	\$10 / t	Basalt fines as quarry by-product.
USD_PER_MIN_PER_T	\$0.04 / min / t	30-t truck at \$80 / h operating cost.
EFFECTIVE_KM_PER_MIN	0.5 ($\approx$ 30 km/h)	Effective speed for distance approximation.
MAX_HAUL_KM	1000 km	Cap beyond which lifecycle emissions erase the credit.

Where to look: `erw/erw-basalt-access.R`, `erw/erw-build-basalt-mask.R`, `erw/erw-fetch-friction-surface.R`, `docs/erw-parameters.md` §4.

B5. Energy — erw-energy-country — REAL

Per-country industrial-electricity tariff (\$/kWh) and grid carbon intensity (kg CO₂/kWh) rasterised onto the working grid. Drives both the grinding-cost term and the grinding-LCA deduction inside erw-7. Falls back to flat \$0.10 / kWh + 0.60 kg / kWh when missing.

Where to look: `erw/erw-energy-country.R`, `country_energy_table.csv`.

B6. Crop response — EcoCrop arm (erw-4, erw-5) — REAL; Bayesian arm (erw-yield-fit, erw-yield-predict) — NOT RUN

EcoCrop arm runs: per-crop pH and acidity-saturation tolerance tables (23 crops $\times$ 2 responses) feed a Recocrop relative-yield surface clamped to [0.2, 1]. Output: `ecocrop_f/{hp,ph}_crop_suitability_<crop>`

Bayesian arm is unfit: needs brms + Stan + the ~14-row trial dataset. When fit, erw-7 prefers its posterior medians and falls back to EcoCrop where the Bayesian surface is NA.

Where to look: erw/erw-4-crop-parameters.R, erw/erw-5-crop-yield-loss-function.R, erw/erw-yield-fit.R, erw/erw-yield-predict.R, erw/erw-trial-data.csv.

B7. Profitability — erw-7 — NOT RUN

Cost side: quarry + grinding + transport + spreading. Credit side: $\text{CDR} \times p_{\text{CO}_2}$ — MRV. Yield side: $\Delta y \times p_{\text{crop}}$. Three regimes (year-1, NPV @ 10%, equilibrium) with a 30 / 25 / 20 / 15 / 10 % CDR phasing over five years.

Key constants

Name	Default	Notes
SPREADING_USD_T	\$8 / t	On-farm spreading cost.
ELECTRICITY_USD_KWH_FB	\$0.10 / kWh	Fallback when country raster missing.
GRID_CI_KG_PER_KWH_FB	0.60 kg / kWh	Fallback when country raster missing.
TRUCK_KGC02_PER_T_KM	0.12	Diesel heavy truck.
CARBON_PRICE_USD_T	\$150 / tCO ₂	Mid of 2024–25 ERW credit market.
MRV_COST_USD_T_CO2	\$30 / tCO ₂	Sampling + lab + verification + registry.
DISCOUNT_RATE_PCT	10 %	NPV regime.
CDR_PHASING	30 / 25 / 20 / 15 / 10 % over 5 yr	Time-resolved CDR realisation.

Where to look: erw/erw-7-profitability-function.R, docs/erw-parameters.md §5.

B8. Supply ranking — erw-supply — NOT RUN

Pixels ranked descending by gross margin per tonne basalt — the LP dual on the basalt-supply constraint. Produces supply curves and deployment masks at 1 / 5 / 10 / 25 / 50 / 100 / 250 / 500 Mt yr⁻¹ caps, plus an unconstrained reference.

Where to look: erw/erw-supply.R.

B9. Sensitivity — erw-8 — NOT RUN

Five sweeps: (i) crop × basalt × carbon-price grid; (ii) yield uplift 1.0 – 2.5×; (iii) MRV 0 – 80 \$/tCO₂; (iv) grain size 10 / 30 / 50 / 100 / 200 / 500 μm; (v) allocation rule (LiTAS-targeted + uniform 10 / 20 / 50). All sweeps run across the three regimes and 23 crops.

Where to look: erw/erw-8-sensitivity-analysis.R.

7. Outputs and indicator maps

Twelve indicator maps are rendered into docs/maps/ by erw/erw-plot-maps.R. The HTML page embeds them in a gallery alongside the conceptual-framework and pipeline-architecture figures.

#	Map	What it shows	Block
01	01-soil-ph.png	SoilGrids depth-weighted topsoil pH, cropland-masked	B1
02	02-basalt-rate-merlos.png	LiTAS year-1 basalt application rate (t / ha)	B2
03	03-cdr-per-t-basalt.png	Per-tonne CDR efficiency (t CO ₂ / t basalt)	B3
04	04-cdr-yield-merlos.png	Per-ha cumulative CDR (t CO ₂ / ha)	B3
05	05-cdr-climate-factor.png	$f(\text{MAT}, \text{MAP})$ scaling	B3
06	06-pedogenic-fraction.png	Aridity-band pedogenic-C deduction	B3
07	07-electricity-price.png	Country industrial-electricity tariff (/kWh) B5 08 `08 – grid – CI.png Countrygridcarbonintensity(kgCO <sub>2</sub> /kWh) B5  basalt – sources.png GLiMvb+ pbsourcemask B4  10 `10– basalt – traveltime.png Traveltimetonearestsource(min) B4 1 basalt – delivered – price.png Deliveredfeedstockprice(/ t)	B4
12	12-feedstock-class.png	Source mask split by GLiM class	B4

Add a row when a new map is rendered; remove when one is dropped.

8. Parameters and constants

See docs/erw-parameters.md for the full catalogue with citations, source URLs and DOIs. The blocks in §6 above repeat only the constants that materially set the headline result; the parameters doc is authoritative for everything else.

Quick reference of the six **decision variables** (the only knobs swept by erw-8):

Decision variable	Default	Where set	Sweep range
CaO_frac, MgO_frac (feedstock)	0.10 / 0.07	erw-3	not currently swept
grain_size_um	100 µm	erw-3	10 / 30 / 50 / 100 / 200 / 500 µm
allocation	LiTAS-targeted	erw-7, erw-8	+ uniform 10 / 20 / 50
CARBON_PRICE_USD_T	\$150 / tCO ₂	erw-7, erw-8	swept jointly with crop + basalt price
MRV_COST_USD_T_CO2	\$30 / tCO ₂	erw-7, erw-8	0 – 80 \$ / tCO ₂
DISCOUNT_RATE_PCT	10 %	erw-7	NPV regime only

9. Validation and known issues

What has been sanity-checked

- B1 pH map matches the qualitative expectation for SSA (acidic in humid tropics, alkaline in Sahel and parts of southern Africa).
- B3 CDR-yield map is strongest in the humid tropics (Guinea, Congo basin) and suppressed in the Sahel — the expected pattern given climate scaling and the aridity-driven pedogenic-C deduction.
- B4 delivered-price map shows the expected geography — cheap near basalt outcrops (Ethiopian highlands, East African Rift, Cameroon volcanic line, Madagascar, southern Africa), capped at \$50 / t across West Africa and the Sahel.
- B4 source-mask polygon counts match the GLiM v1 catalogue (5,266 vb + 1,189 pb in SSA).

Open issues

Issue	Severity	Block	Mitigation
Bayesian yield arm not fit	high (limits scenario realism)	B6	run <code>erw-yield-fit.R</code> once <code>brms</code> + <code>Stan</code> are installed; the trial dataset is the bottleneck, not the code
<code>erw-7</code> profitability not run	high (no headline output yet)	B7	all inputs are in place; one run suffices
<code>erw-supply</code> , <code>erw-8</code> not run	medium	B8, B9	both depend on B7
Industrial by-products not modelled	medium (regional realism)	B4	requires a separate inventory of steel mills, cement plants, mine tailings
Ultramafic outcrops excluded from feedstock	low (intentional)	B4	mark them explicitly in a future “rejected feedstock” overlay
MRV cost is constant \$30 / tCO ₂ regardless of scale or protocol	low	B7	sweep already covers 0 – 80 \$; consider scale-dependence in a future revision
Pedogenic-C deduction uses MAP only, not full AI	low	B3	switch to AI = MAP / PET when a PET surface is available
14-row trial dataset is prior-dominated	medium	B6 (Bayesian arm)	append trials as new evidence appears (Beerling group, InPlanet, Flux trials)
Single-currency model	low	all	USD throughout; not a problem for ex-ante analysis

10. How to reproduce

10.1 Environment

R 4.5+, with the `geodata`, `terra`, `sf`, `here`, `brms`, `malariaAtlas` packages

Rscript `erw/erw-0-install-packages.R`

LaTeX is needed only for the framework + pipeline figures (`pdflatex` + `pdftocairo`).

10.2 Run order

1. inputs and working grid

```
Rscript erw/erw-1-layers-soilgrids.R
```

```
Rscript erw/erw-2-layers-spam.R
```

2. basalt requirements

```
Rscript erw/erw-3-basalt-requirements.R
```

3. crop response

```
Rscript erw/erw-4-crop-parameters.R
```

```
Rscript erw/erw-5-crop-yield-loss-function.R
```

4. CDR + cost surfaces

```
Rscript erw/erw-9-cdr-yield.R
```

```
Rscript erw/erw-energy-country.R
```

```
Rscript erw/erw-fetch-friction-surface.R # one-off – pulls MAP friction
```

```
Rscript erw/erw-basalt-access.R
```

5. (optional) Bayesian yield arm

```
Rscript erw/erw-yield-fit.R
```

```
Rscript erw/erw-yield-predict.R
```

6. profitability, supply, sensitivity

```
Rscript erw/erw-7-profitability-function.R
```

```
Rscript erw/erw-supply.R
```

```
Rscript erw/erw-8-sensitivity-analysis.R
```

7. render maps and rebuild figures

```
Rscript erw/erw-plot-maps.R
```

```
pdflatex docs/erw-pipeline-flow.tex && pdftocairo -svg docs/erw-pipeline-flow.pdf
```

```
pdflatex docs/erw-conceptual-framework.tex && pdftocairo -svg docs/erw-conceptual-framework
```

10.3 Expected runtimes (Apple Silicon, single core)

Stage	Time
erw-1, erw-2 (downloads + resampling)	~30 – 60 min first time, cached afterwards
erw-3, erw-9	a few minutes each
erw-fetch-friction-surface	5 – 10 min (one-off download)
erw-basalt-access (costDist)	~10 min
erw-7, erw-8	minutes – hours depending on sweep depth
erw-plot-maps (12 maps)	~15 min

11. Gaps and roadmap

Near-term (within current month)

1. Fit the Bayesian yield arm (erw-yield-fit.R) once brms + Stan are installed locally.

2. Run erw-7 end-to-end to produce the 276-raster profitability stack (economics_erw/{allocation}/{crop
3. Run erw-supply and erw-8 and render their headline figures.
4. Add indicator maps for the supply curves and the year-1 / NPV / equilibrium profitability comparison.

Medium-term

5. **Industrial by-product feedstock layer.** Inventory of steel mills (GEM Steel Plant Tracker), cement-clinker plants (Global Cement Directory), and active mine tailings (USGS / national geological surveys). Add a point-source feedstock layer and feedstock- chemistry table; rerun B4 with the union of mafic outcrops and by-product points.
6. **Cross-country MRV cost differentiation.** Some jurisdictions (Brazil, Kenya) have simpler smallholder-aggregator pathways; the flat \$30 / tCO₂ assumption over-estimates cost for those.
7. **Aridity index using AI = MAP / PET** instead of MAP-only for the pedogenic-C deduction. Requires a PET surface (CGIAR-CSI or computed from WorldClim).

Long-term

8. **Multi-feedstock decision rule.** Currently feedstock chemistry is fixed; the user chooses one. Generalise to a per-pixel cheapest-eligible-feedstock rule given a chemistry inventory.
9. **Time-resolved deployment scenarios.** Year-by-year deployment under supply caps, carbon-price paths, and yield-learning curves.
10. **Smallholder-aggregator economic model.** Costs and revenues at the aggregator level (transport, MRV pooling, finance) rather than the per-pixel “farmer” level.

12. References

The full bibliography is in docs/erw-review.md. The most load-bearing citations are:

- Beerling, D. J. et al. (2020). *Potential for large-scale CO₂ removal via enhanced rock weathering with croplands*. **Nature** 583, 242 – 248.
 - Strefler, J. et al. (2018). *Potential and costs of carbon dioxide removal by enhanced weathering of rocks*. **ERL** 13, 034010.
 - Lewis, A. L. et al. (2021). *Effects of mineralogy, chemistry and climatic regime on rates of basalt weathering*. (Reference for the reactive-fraction calibration used in erw-3.)
 - Hartmann, J. & Moosdorf, N. (2012). *The new global lithological map database GLiM*. PANGAEA doi:10.1594/PANGAEA.788537.
 - Weiss, D. J. et al. (2018, 2020). *Global maps of travel time to healthcare facilities*. Malaria Atlas Project — friction surface 2019.
 - Aramburu-Merlos, F. et al. (2024). *LiTAS: lime-targeted application strategy for tropical acid soils*. (Reference for the LiTAS lime-equivalence recipe used in erw-3.)
 - Kanzaki, Y. et al. (2024). *Soil-pH-driven controls on enhanced-weathering CDR rates*. **Nature**.
 - Beerling group, US Corn Belt field trials (2024 PNAS).
 - Flux × UNCCD Kisumu County trial (2024).
 - Amann, T. et al. (2025). *Swiss vineyard ERW trial: low CDR rates in cool-alkaline systems*. **Environmental Science & Technology**.
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This document is the canonical state of the model. When you change a script, a constant, a data source, or an output, edit the relevant block under §6 and add a dated entry to §1. Anything that is not in this document — even if it lives in the code — is not part of the model's documented behaviour.